

Title: Socioeconomic and Health-System Correlates of Cesarean Delivery in Bangladesh

Manuscript Status Note:

This research is currently under peer review. To maintain confidentiality during the review process, detailed quantitative results, regression estimates, and specific performance metrics have been omitted. This summary focuses on the research context, methodology, analytical framework, and policy-simulation approach.

1. Research Context

Cesarean delivery has increased substantially in Bangladesh alongside the expansion of institutional maternal healthcare. While cesarean delivery is an essential life-saving intervention when medically indicated, its increasing use raises concerns regarding clinical appropriateness, household financial burden, and socioeconomic inequalities in maternal healthcare.

Using nationally representative **Bangladesh Demographic and Health Survey (BDHS) 2022** data, this study examines how socioeconomic, maternal, reproductive, healthcare-utilization, geographic, and delivery-context factors are associated with cesarean delivery.

The study combines **survey-weighted statistical inference, robustness analysis, machine learning, Explainable AI, and model-based counterfactual policy simulation** within an integrated analytical framework.

2. Methodological Framework

Analytical Stage	What Was Done	Why It Was Done
Data Preparation & Survey Design	Data cleaning, variable construction, exploratory analysis, and incorporation of BDHS survey weights, stratification, and PSU clustering	To appropriately account for the complex survey design and support population-level inference
Survey-Weighted Inference	Survey-weighted descriptive analysis and binomial GEE with logit link and robust sandwich standard errors	To estimate population-level associations while accounting for within-cluster correlation
Robustness & Sensitivity Analysis	Propensity-score-based IPTW, covariate-adjusted weighted GEE, and E-value analysis	To assess the robustness of key associations and their sensitivity to potential unmeasured confounding

Feature Selection	Recursive Feature Elimination (RFE), Random Forest feature importance, and Mutual Information	To identify informative predictors and reduce unnecessary features
Predictive Modeling	Optimized XGBoost, LightGBM, CatBoost, and Extra Trees models with a stacking ensemble	To capture complex and nonlinear predictive patterns and combine complementary model strengths
Model Validation & Stress Testing	Stratified cross-validation and controlled Gaussian-noise testing	To evaluate model stability and sensitivity to controlled data perturbations
Explainable AI	SHAP for global and local interpretation, with partial dependence and Friedman H-statistics	To understand predictor contributions and investigate nonlinear relationships and interactions
Counterfactual Policy Simulation	One-at-a-time modifications of selected healthcare-utilization and delivery-context inputs while holding other covariates constant	To examine how predicted probabilities respond under alternative modeled scenarios

3. Research Contribution

The main contribution of this study is the integration of **population-level statistical inference and modern predictive analytics** in a single maternal-health research framework.

By combining robust statistical analysis with machine learning, Explainable AI, and scenario-based simulation, the study provides a multidimensional perspective on the roles of **healthcare utilization, delivery settings, socioeconomic position, and geographic context** in cesarean delivery.

Importantly, statistical associations, predictive patterns, and simulated scenarios are interpreted as distinct analytical components rather than as interchangeable evidence of causality.

4. Future Direction

The framework provides a foundation for future research incorporating richer **clinical, provider-level, and facility-level data** to better distinguish medically indicated cesarean delivery from potentially avoidable use and to strengthen predictive validation and policy-oriented analysis.

This summary intentionally omits detailed quantitative findings and specific statistical estimates while presenting the overall research design and methodological contribution.